

Go Data Science & AI Governance Roadmap

This roadmap combines data science learning, Go engineering, and AI governance. The goal is to understand algorithms from first principles while building skills that can be applied directly to production AI systems and startup development.

Track 1: Core Statistics

- Mean, Median, Mode
- Variance and Standard Deviation
- Covariance and Correlation
- Probability Fundamentals
- Bayes Theorem
- Confidence Intervals
- Hypothesis Testing
- A/B Testing Engine

Track 2: Machine Learning From Scratch

- Linear Regression
- Gradient Descent
- R^2 and Adjusted R^2
- Logistic Regression
- Decision Trees
- K-Nearest Neighbors
- Naive Bayes
- K-Means Clustering

Track 3: Data Engineering

- CSV Processing
- Data Cleaning Pipelines
- Feature Engineering Framework
- Mini DataFrame Implementation
- Streaming Data Processing
- Analytics Aggregation Services

Track 4: AI Governance & Compliance

- PII Detection
- Prompt and Response Auditing
- Risk Scoring Engine
- Policy Enforcement Engine

- Agent Monitoring
- Hallucination Detection
- Explainability Framework
- Compliance Reporting

Track 5: Production Skills

- Use Gonnun for numerical computing
- Use GoLearn for ML algorithms
- Explore Gorgonia for deep learning
- Build APIs around models
- Deploy and monitor systems
- Design scalable AI services

Suggested Repository Structure

```
go-first-principles-ml/  
  statistics/  
  probability/  
  linear_regression/  
  logistic_regression/  
  decision_trees/  
  knn/  
  naive_bayes/  
  kmeans/  
  ab_testing/  
  pii_detection/  
  risk_scoring/  
  agent_monitoring/  
  hallucination_detection/  
  analytics/
```